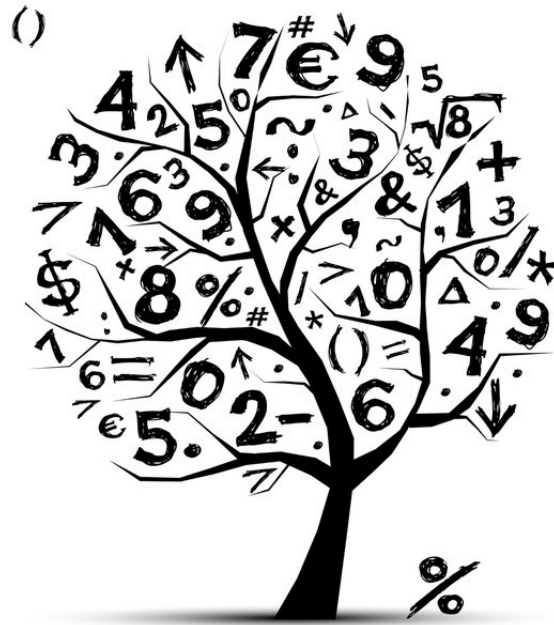


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Exercises: eigenvalues and eigenvectors



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Exercise 1

Compute the eigenvalues of the matrix

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$$

and one eigenvector associated with each eigenvalue.

The characteristic equation is

$$\det(A - \lambda I) = \det \begin{pmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{pmatrix} = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5) = 0,$$

hence $\lambda_1 = 2$ and $\lambda_2 = 5$.

For $\lambda_1 = 2$:

$$(A - 2I)\mathbf{v} = \mathbf{0} \iff \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \mathbf{v} = \mathbf{0} \iff 2v_1 + v_2 = 0, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

For $\lambda_2 = 5$:

$$(A - 5I)\mathbf{v} = \mathbf{0} \iff \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \mathbf{v} = \mathbf{0} \iff v_1 = v_2, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Check: $A\mathbf{v}_1 = (4 - 2, 2 - 6)' = (2, -4)' = 2\mathbf{v}_1$ and $A\mathbf{v}_2 = (5, 5)' = 5\mathbf{v}_2$.

Exercise 2

Compute the eigenvalues and one eigenvector for each eigenvalue of the symmetric matrix

$$Q = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}.$$

Is Q positive definite?

The characteristic equation is

$$\det(\mathbf{Q} - \lambda \mathbf{I}) = (5 - \lambda)(2 - \lambda) - 4 = \lambda^2 - 7\lambda + 6 = (\lambda - 1)(\lambda - 6) = 0,$$

hence $\lambda_1 = 1$ and $\lambda_2 = 6$.

For $\lambda_1 = 1$:

$$\begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} \mathbf{v} = \mathbf{0} \iff 2v_1 + v_2 = 0, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

For $\lambda_2 = 6$:

$$\begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix} \mathbf{v} = \mathbf{0} \iff v_1 = 2v_2, \quad \mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Since $\mathbf{Q}$ is symmetric and both eigenvalues are strictly positive, $\mathbf{Q}$ is **positive definite**. (Note also that $\mathbf{v}_1' \mathbf{v}_2 = 2 - 2 = 0$.)

Exercise 3

Without computing any determinant, find the eigenvalues of

$$\mathbf{U} = \begin{pmatrix} 3 & 1 & -2 \\ 0 & -1 & 4 \\ 0 & 0 & 2 \end{pmatrix}.$$

Then find one eigenvector for each eigenvalue and check the trace and the determinant.

$\mathbf{U}$ is upper triangular, hence its eigenvalues are its diagonal entries:

$$\lambda_1 = 3, \quad \lambda_2 = -1, \quad \lambda_3 = 2.$$

- $\lambda_1 = 3$: $(\mathbf{U} - 3\mathbf{I})\mathbf{v} = \mathbf{0}$ gives $v_2 - 2v_3 = 0$, $-4v_2 + 4v_3 = 0$, $-v_3 = 0$, hence $v_2 = v_3 = 0$ and $\mathbf{v}_1 = (1, 0, 0)'$.
- $\lambda_2 = -1$:

$$(\mathbf{U} + \mathbf{I})\mathbf{v} = \begin{pmatrix} 4 & 1 & -2 \\ 0 & 0 & 4 \\ 0 & 0 & 3 \end{pmatrix} \mathbf{v} = \mathbf{0} \implies v_3 = 0, \quad 4v_1 + v_2 = 0, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -4 \\ 0 \end{pmatrix}.$$

- $\lambda_3 = 2$:

$$(\mathbf{U} - 2\mathbf{I})\mathbf{v} = \begin{pmatrix} 1 & 1 & -2 \\ 0 & -3 & 4 \\ 0 & 0 & 0 \end{pmatrix} \mathbf{v} = \mathbf{0} \implies v_2 = \frac{4}{3}v_3, \quad v_1 = -v_2 + 2v_3 = \frac{2}{3}v_3.$$

Choosing $v_3 = 3$: $\mathbf{v}_3 = (2, 4, 3)'$.

Check: $\text{tr}(\mathbf{U}) = 3 - 1 + 2 = 4 = \lambda_1 + \lambda_2 + \lambda_3$ and $\det(\mathbf{U}) = 3 \cdot (-1) \cdot 2 = -6 = \lambda_1 \lambda_2 \lambda_3$.

Exercise 4

Compute the characteristic polynomial, the eigenvalues and one eigenvector for each eigenvalue of

$$\mathbf{Q} = \begin{pmatrix} -1 & 1 & 0 \\ 1 & 2 & 3 \\ 0 & 3 & -1 \end{pmatrix}.$$

Classify $\mathbf{Q}$ (definite, semidefinite or indefinite).

Expanding along the first row:

$$\begin{aligned} \det(\mathbf{Q} - \lambda \mathbf{I}) &= \det \begin{pmatrix} -1-\lambda & 1 & 0 \\ 1 & 2-\lambda & 3 \\ 0 & 3 & -1-\lambda \end{pmatrix} \\ &= (-1-\lambda)[(2-\lambda)(-1-\lambda)-9] - 1 \cdot [1 \cdot (-1-\lambda) - 3 \cdot 0] \\ &= (-1-\lambda)[(2-\lambda)(-1-\lambda)-9-1] = (-1-\lambda)(\lambda^2 - \lambda - 12) \\ &= (-1-\lambda)(\lambda-4)(\lambda+3). \end{aligned}$$

The eigenvalues are $\lambda_1 = -1$, $\lambda_2 = 4$, $\lambda_3 = -3$.

- $\lambda_1 = -1$: $(\mathbf{Q} + \mathbf{I})\mathbf{v} = \mathbf{0}$ with $\mathbf{Q} + \mathbf{I} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 3 & 3 \\ 0 & 3 & 0 \end{pmatrix}$ gives $v_2 = 0$ and $v_1 + 3v_3 = 0$:
 $\mathbf{v}_1 = (3, 0, -1)'$.

- $\lambda_2 = 4$: $\mathbf{Q} - 4\mathbf{I} = \begin{pmatrix} -5 & 1 & 0 \\ 1 & -2 & 3 \\ 0 & 3 & -5 \end{pmatrix}$ gives $v_2 = 5v_1$, $v_3 = \frac{3}{5}v_2 = 3v_1$ (and $v_1 - 10v_1 + 9v_1 = 0$): $\mathbf{v}_2 = (1, 5, 3)'$.

- $\lambda_3 = -3$: $\mathbf{Q} + 3\mathbf{I} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 5 & 3 \\ 0 & 3 & 2 \end{pmatrix}$ gives $v_2 = -2v_1$, $v_3 = -\frac{3}{2}v_2 = 3v_1$ (and $v_1 - 10v_1 + 9v_1 = 0$): $\mathbf{v}_3 = (1, -2, 3)'$.

Check: $\text{tr}(\mathbf{Q}) = -1 + 2 - 1 = 0 = -1 + 4 - 3$ and $\det(\mathbf{Q}) = -1 \cdot (-2 - 9) - 1 \cdot (-1 - 0) = 12 = (-1) \cdot 4 \cdot (-3)$.

$\mathbf{Q}$ is symmetric with a positive and a negative eigenvalue, hence it is **indefinite**.

Exercise 5

The matrix

$$\mathbf{Q} = \begin{pmatrix} 4 & -1 & 1 \\ -1 & 4 & -1 \\ 1 & -1 & 4 \end{pmatrix}$$

has the eigenvalue $\lambda = 3$ with multiplicity 2. Using the trace, find the third eigenvalue; then check the result with the determinant and find an eigenvector for it.

The eigenvalues, counted with multiplicity, are $\lambda_1 = \lambda_2 = 3$ and λ_3 . Since the trace is the sum of the eigenvalues:

$$\text{tr}(\mathbf{Q}) = 4 + 4 + 4 = 12 = 3 + 3 + \lambda_3 \implies \lambda_3 = 6.$$

Check with the determinant (Laplace expansion along the first row):

$$\det(\mathbf{Q}) = 4(16 - 1) - (-1)(-4 + 1) + 1(1 - 4) = 60 - 3 - 3 = 54 = 3 \cdot 3 \cdot 6. \quad \checkmark$$

Eigenvector for $\lambda_3 = 6$:

$$(\mathbf{Q} - 6\mathbf{I})\mathbf{v} = \begin{pmatrix} -2 & -1 & 1 \\ -1 & -2 & -1 \\ 1 & -1 & -2 \end{pmatrix} \mathbf{v} = \mathbf{0}.$$

Adding the first two equations: $-3v_1 - 3v_2 = 0$, i.e., $v_2 = -v_1$; then from the first equation $v_3 = 2v_1 + v_2 = v_1$. Hence $\mathbf{v} = (1, -1, 1)'$; indeed $\mathbf{Q}\mathbf{v} = (4 + 1 + 1, -1 - 4 - 1, 1 + 1 + 4)' = (6, -6, 6)' = 6\mathbf{v}$.

Exercise 6

Using the eigenvalues, classify the following symmetric matrices:

$$\text{a) } \mathbf{Q}_1 = \begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix}, \quad \text{b) } \mathbf{Q}_2 = \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}, \quad \text{c) } \mathbf{Q}_3 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}.$$

- a) $(-3 - \lambda)^2 - 1 = \lambda^2 + 6\lambda + 8 = (\lambda + 2)(\lambda + 4)$: the eigenvalues are -2 and -4 , both strictly negative, hence $\mathbf{Q}_1$ is **negative definite**.
- b) $(1 - \lambda)^2 - 9 = \lambda^2 - 2\lambda - 8 = (\lambda - 4)(\lambda + 2)$: the eigenvalues are 4 and -2 , hence $\mathbf{Q}_2$ is **indefinite**.
- c) $(2 - \lambda)^2 - 4 = \lambda^2 - 4\lambda = \lambda(\lambda - 4)$: the eigenvalues are 0 and 4 , both non-negative, hence $\mathbf{Q}_3$ is **positive semidefinite** (but not positive definite, since 0 is an eigenvalue).

Exercise 7

Using Sylvester's criterion, prove that the matrix

$$\mathbf{Q} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$$

is positive definite.

$\mathbf{Q}$ is symmetric and its leading principal minors are

$$\Delta_1 = 2 > 0, \quad \Delta_2 = \det \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = 4 - 1 = 3 > 0,$$

$$\Delta_3 = \det(\mathbf{Q}) = 2(4 - 1) - (-1)(-2 - 0) + 0 = 6 - 2 = 4 > 0.$$

All the leading principal minors are strictly positive, hence $\mathbf{Q}$ is **positive definite**. (Here Sylvester's criterion is convenient: the eigenvalues of $\mathbf{Q}$ are 2 and $2 \pm \sqrt{2}$, which are not integers.)

Exercise 8

Using Sylvester's criterion, classify

$$\mathbf{Q} = \begin{pmatrix} -2 & 1 & 0 \\ 1 & -3 & 1 \\ 0 & 1 & -2 \end{pmatrix}.$$

$\mathbf{Q}$ is symmetric and its leading principal minors are

$$\Delta_1 = -2 < 0, \quad \Delta_2 = \det \begin{pmatrix} -2 & 1 \\ 1 & -3 \end{pmatrix} = 6 - 1 = 5 > 0,$$

$$\Delta_3 = \det(\mathbf{Q}) = -2(6 - 1) - 1(-2 - 0) = -10 + 2 = -8 < 0.$$

The leading principal minors alternate in sign starting with a negative one, hence $\mathbf{Q}$ is **negative definite**. (Indeed, its eigenvalues are -1 , -2 and -4 .)

Exercise 9

Using the principal minors, decide whether the following symmetric matrices are positive semidefinite:

$$\text{a) } \mathbf{Q}_a = \begin{pmatrix} 2 & -2 & 0 \\ -2 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \text{b) } \mathbf{Q}_b = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

a) The leading principal minors are $\Delta_1 = 2$, $\Delta_2 = 4 - 4 = 0$, $\Delta_3 = \det(\mathbf{Q}_a) = 0$: $\mathbf{Q}_a$ is not positive definite. We check **all** principal minors:

- order 1: 2, 2, 1;
- order 2: $\det \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix} = 0$, $\det \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = 2$, $\det \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = 2$;
- order 3: $\det(\mathbf{Q}_a) = 0$.

All are ≥ 0 : $\mathbf{Q}_a$ is **positive semidefinite**. Indeed $\mathbf{x}'\mathbf{Q}_a\mathbf{x} = 2(x_1 - x_2)^2 + x_3^2 \geq 0$, and the eigenvalues are 0, 1, 4.

b) The leading principal minors are $\Delta_1 = 1$, $\Delta_2 = 1 - 1 = 0$, $\Delta_3 = \det(\mathbf{Q}_b) = -1 \cdot 0 = 0$: they are all ≥ 0 , but this is **not** enough. The principal minor of order 1 given by the entry $q_{33} = -1$ is negative, hence $\mathbf{Q}_b$ is **not** positive semidefinite: for $\mathbf{x} = (0, 0, 1)'$ we have $\mathbf{x}'\mathbf{Q}_b\mathbf{x} = -1 < 0$. Since for $\mathbf{x} = (1, 0, 0)'$ we have $\mathbf{x}'\mathbf{Q}_b\mathbf{x} = 1 > 0$, the matrix is **indefinite** (its eigenvalues are 2, 0, -1).

Exercise 10

Let $k \in \mathbb{R}$ and

$$\mathbf{Q}(k) = \begin{pmatrix} 2 & 1 & 0 \\ 1 & k & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

For which values of k is $\mathbf{Q}(k)$ positive definite? For which values is it positive semidefinite?

$\mathbf{Q}(k)$ is symmetric for every k . Its leading principal minors are

$$\Delta_1 = 2, \quad \Delta_2 = \det \begin{pmatrix} 2 & 1 \\ 1 & k \end{pmatrix} = 2k - 1, \quad \Delta_3 = 2(2k - 1) - 1 \cdot (2 - 0) = 4k - 4.$$

By Sylvester's criterion, $\mathbf{Q}(k)$ is positive definite if and only if

$$2 > 0, \quad 2k - 1 > 0 \Leftrightarrow k > \frac{1}{2}, \quad 4k - 4 > 0 \Leftrightarrow k > 1,$$

that is, if and only if $k > 1$.

For positive semidefiniteness we need all principal minors ≥ 0 : order 1: 2, k , 2; order 2: $2k - 1$, $\det \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 4$, $2k - 1$; order 3: $4k - 4$. They are all ≥ 0 if and only if $k \geq 0$, $k \geq \frac{1}{2}$ and $k \geq 1$, i.e., if and only if $k \geq 1$. For $k = 1$ the matrix is positive semidefinite but not positive definite: its eigenvalues are 0, 2, 3.